Faithful or Plausible? A Ground-Truth Evaluation of Explainable AI for 3D MRI-Based Alzheimer's Disease Classification

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Abstract—Heatmaps that explain deep-learning models of Alzheimer's disease are commonly validated by showing that they highlight the hippocampus, but such plausibility does not establish faithfulness, and on real cohorts neither can be checked against ground truth. We present a benchmark in which the disease signal is known voxel by voxel: hippocampal atrophy is simulated on 581 healthy T1-weighted scans from the open IXI dataset by a fold-checked displacement field with an exact truth map, and a faint intensity shortcut is optionally planted far from the hippocampus. We evaluate six attribution methods for a 3D ResNet on plausibility, faithfulness and correctness with 5-fold cross-validation. The classifier detected atrophy (AUC 0.992 ± 0.006) and stayed at chance on a resampling control (0.515 ± 0.036). Faithfulness and correctness ranked the methods identically, whereas plausibility exaggerated their differences, and faithfulness remained high on a model that had learned nothing. A no-signal control showed that Grad-CAM and Grad-CAM++ assign 2.7–3.0 times chance relevance to the shortcut's position in a model that never saw it. LRP, Integrated Gradients and Gradient $\times$ Input tracked the model's actual reliance across a sweep of shortcut strength, and in randomisation tests Grad-CAM++ maps were almost unchanged by a random output layer (Spearman $\rho = 0.95$). Evaluating explanations needs ground truth, a reliance test, a no-signal control and randomisation tests.

Index Terms—Explainable AI, Layer-wise relevance propagation, Faithfulness, Alzheimer's disease, Structural MRI, Ground truth, Shortcut learning

I. Introduction

Convolutional networks classify Alzheimer's disease (AD) from structural MRI with high reported accuracy, and attribution maps are increasingly presented as evidence that these models attend to disease-relevant anatomy [1], [2]. The usual argument is visual: relevance concentrates on the hippocampus, which is what a clinician expects. Jacovi and Goldberg [3] distinguish this plausibility from faithfulness, the degree to which an explanation reflects the model's actual reasoning, and argue that the two must be evaluated separately.

For AD the separation matters for three reasons. Models exploit shortcuts such as age, scanner and acquisition artefacts [4]; a faithful explanation of such a model should reveal the shortcut, not the hippocampus. Leakage between visits of the same subject inflates accuracy [5], so explanations of leaky models are uninterpretable. And attribution implementations can be wrong while still producing convincing maps, as we show for LRP in Section IV-A.

The obstacle is ground truth: on a real cohort nobody knows which voxels a model should use, so a plausible map cannot be shown to be unfaithful. We remove this obstacle by constructing the disease signal ourselves on healthy brains. Our contributions are:

- a ground-truth benchmark for 3D MRI explanations built on open IXI data, with simulated hippocampal atrophy, an exact voxel-wise truth map, a planted shortcut with controlled label correlation and a resampling control;
- a three-axis evaluation (plausibility, faithfulness and correctness, measured on the same maps) of six attribution methods on 581 subjects with 5-fold cross-validation;
- controls that change the conclusions: a reliance test independent of any explanation, a sweep over shortcut strength, a no-signal location control, which reveals that Grad-CAM and Grad-CAM++ appear to detect a shortcut purely because of where it is placed, and model- and data-randomisation tests, which Grad-CAM++ fails;
- implementation checks for LRP in residual 3D networks, including a sign error that only a conservation test caught.

II. Related Work

Attribution methods. LRP [6], [7] redistributes the output score backwards under a conservation constraint, with composite rules recommended for deep networks. Grad-CAM [8] weights the final convolutional feature maps by pooled gradients and is upsampled from a coarse grid in 3D. Integrated Gradients [9] integrates gradients along a path from a baseline. Occlusion and Gradient $\times$ Input serve as model-agnostic and first-order baselines.

Explanations for AD. Böhle et al. [1] applied LRP to a 3D CNN on ADNI and found relevance in the hippocampus and temporal lobe. Dyrba et al. [2] evaluated relevance maps for AD, including a simulated-atrophy check in which intensity was reduced in a cube. Such studies establish plausibility; faithfulness is rarely measured on the same maps.

Evaluating explanations. Insertion and deletion curves [10] measure how the score changes as features are added or removed in order of attribution. ROAR [11] retrains after removal, which is costly in 3D, and ROAD [12] reduces leakage from the removal pattern. Sanity checks [13] test whether maps depend on the model at all, and Arun et al. [14] found that many saliency methods fail localisation and randomisation tests on chest radiographs. Biophysical simulators generate MRI with known volume change [15]; we use a lightweight displacement-field model chosen for an exact, verifiable truth map.

III. Methods

A. Data and Preprocessing

We use the IXI dataset [16]: T1-weighted scans of healthy adults from three London sites, released under CC BY-SA 3.0. Each scan is bias-corrected (N4), affinely registered head-to-head to the ICBM152 2009 template on the 2 mm MNI grid, brain-extracted with the template mask dilated by two voxels, resampled to 96^3 and z-scored within the brain. A bilateral hippocampus mask from the Harvard-Oxford atlas is resampled identically and its placement was checked visually on scans from two sites. All 581 scans were processed without failure. The cohort is 48.7 ± 16.5 years old (range 20–86), 313 female and 253 male, from Guys (322), Hammersmith (185) and the Institute of Psychiatry (74).

B. Simulated Atrophy with Exact Ground Truth

Let c_k be the centroid of hippocampal component k (left, right), $q_k(x)$ the nearest point of component k to voxel x and $d_k(x)$ the distance to it. We define the backward displacement

$$u(x) = s \sum_k w_k(x) \big(q_k(x) - c_k\big), \tag{1}$$

with $w_k = 1$ inside the component and $w_k = \exp(-d_k^2/2\sigma^2)$ outside, and resample $\mathrm{out}(x) = \mathrm{in}(x + u(x))$. Inside each component the field is a uniform shrink by $1/(1+s)$, so a star-shaped structure loses $1-(1+s)^{-3}$ of its volume; outside, the boundary displacement fades with distance. The scale s is set from a target loss drawn uniformly from 15–30%. The ground-truth map is $|\det(I+\nabla u)-1|$ thresholded at 0.02, and any deformation with a non-positive Jacobian determinant is rejected. Across the 291 positives the achieved loss, measured by partial-volume integration of the warped mask, was 21.2% on average (range 13.8–28.7%) and the smallest Jacobian determinant was 0.605.

C. Resampling Control, Shortcut and Conditions

Warping interpolates and smooths. So that a classifier cannot detect interpolation instead of atrophy, every scan, positive or control, is resampled once through a random smooth sub-voxel jitter field (maximum 0.5 voxel), to which the atrophy field is added for positives. The shortcut is a Gaussian intensity bump (amplitude 0.3 in z-score units, $\sigma = 4$ voxels) truncated at 5% of its peak, so that no voxel outside its ground-truth support changes, and placed at the brain voxel farthest from the hippocampus. Labels are balanced within site and across age. We build a resampling control (labels assigned, no signal), condition A (atrophy only), condition C (shortcut only) and condition B (atrophy, with the shortcut in exactly a fraction p of positives and $1-p$ of negatives, $p \in \{0.7, 0.9, 1.0\}$, and at $p = 0.9$ also with the bump's amplitude raised to 1.0). Unless stated otherwise, B denotes $p = 0.9$ at amplitude 0.3. For B and C a flipped test set moves the shortcut to the opposite class, which measures reliance without any explanation.

D. Classifier and Attribution Methods

A 3D ResNet-10 at half width is trained with subject-level 5-fold cross-validation (one scan per subject; a split-integrity check raises an error on any overlap), class-weighted loss, mixed precision and early stopping on validation AUC. All methods explain the predicted class. LRP uses a composite of $z^{\mathcal{B}}$ (input layer), $\alpha_1\beta_0$ (lower layers) and ε (upper layers); residual blocks are propagated as units, and BatchNorm is folded into the preceding convolutions. Occlusion replaces a 16^3 window, slid at stride 8, by zero (the brain's mean after z-scoring) and records the drop in class probability; Integrated Gradients uses 32 steps from a Gaussian-blurred copy of the scan ($\sigma = 5$ voxels).

E. Evaluation Axes

Plausibility is the share of positive relevance inside the hippocampus mask divided by the mask's share of brain volume (1.0 = no concentration). Faithfulness is insertion AUC minus deletion AUC over 8^3 patches with a shared blurred baseline. Correctness is the AUPRC of the map against a ground-truth support, divided by its chance level (the support's share of the brain), so that 1.0 means no better than a random map. Every subject is explained once by every method and by a random map; 95% confidence intervals are percentile bootstraps over subjects (2,000 resamples). Hypotheses fixed before the results: H1 plausibility and faithfulness are weakly correlated per volume ($|\rho| < 0.3$); H2 on a shortcut-reliant model at least one method keeps high hippocampal plausibility; H3 Grad-CAM scores higher on plausibility than on correctness, LRP the reverse; H4 the methods differ in how far their maps decorrelate under weight randomisation; H5 on condition A, faithfulness and correctness rank the methods alike.

IV. Results

A. Implementation Validity

The conservation test exposed a sign error in the LRP $z^{\mathcal{B}}$ input rule. The rule computes $z = \mathrm{conv}(a, W) -$

TABLE I
Implementation checks on synthetic data.

Check	Before fix	After fix
LRP conservation error	1.97	1.3×10^{-5}
LRP faithfulness (random: −0.007)	−0.014	+0.035
BatchNorm fold, max output diff.	–	9.8×10^{-7}
Atrophy, 25% target (phantom)	13.9%	22.9%
Shortcut voxels outside its GT	> 0	0

$\mathrm{conv}(l, W^+) - \mathrm{conv}(h, W^-)$; the gradients with respect to the bounds l and h already carry the minus sign, so the relevance must be $a\nabla a + l\nabla l + h\nabla h$. The implementation subtracted the bound terms, and input relevance summed to roughly minus the logit, yet the heatmaps remained visually plausible (Table I).

B. Learned Signal and Reliance

The classifier learned simulated atrophy almost perfectly and the faint shortcut less reliably (Table II). On the resampling control the AUC stayed at chance (every fold between 0.47 and 0.56), so the shared warp does not leak the label. Moving the shortcut to the opposite class reduced the shortcut-only model's accuracy from 0.747 to 0.253 and shifted its logit by 6.46 towards the class the bump indicates. On condition B, 0.4% of decisions changed and the logit moved by 0.10, 1.5% of the shortcut-only model's shift: that model uses atrophy and barely uses the shortcut, although the shortcut appeared in 90% of matching training scans.

C. Three Axes on Atrophy

All six methods exceeded the random map on faithfulness (Table III). Ranked by faithfulness and by correctness, the methods fall in the same order: Occlusion, LRP, Integrated Gradients, Gradient × Input, Grad-CAM, Grad-CAM++ (Spearman $\rho = 1.0$, exact permutation $p = 0.003$), with Integrated Gradients and Gradient × Input tied within their intervals; H5 is supported. Plausibility instead exaggerated the differences: Occlusion concentrated 6.1 times chance relevance on the hippocampus and LRP 2.1 times, against correctness of 3.2 and 1.8. H3 is not supported: Grad-CAM scored lower on plausibility than on correctness (1.14 against 1.32). On the control model, which learned nothing, faithfulness still reached about three-quarters of its atrophy-model value for Occlusion, LRP, Integrated Gradients and Gradient × Input (last column): the score measures how the logit responds to the highlighted voxels, not whether they carry a label-related signal.

D. Shortcuts and a Location Control

On the shortcut-reliant model (C) all methods located the bump well above chance and placed less than chance relevance on the hippocampus (enrichment 0.16–0.81), so low plausibility correctly flagged the problem and H2 is not supported. On model B, which barely uses the shortcut, Grad-CAM, Grad-CAM++ and Occlusion still assigned it 2.5–3.4 times chance relevance. Because the bump sits at the same position in every scan, we explained the atrophy-only model, which never saw a bump, against the same location (Table IV). Grad-CAM and Grad-CAM++ assigned 2.7 and 3.0 times chance relevance there, as much as on model B: their apparent shortcut detection is a property of the position, and even on model C their lift exceeds this baseline by less than a factor of three, against about twenty for LRP. LRP, Integrated Gradients and Gradient × Input retained 1.8–2.7% of their condition-C excess on model B, in proportion to the model's residual dependence (1.5%); Occlusion retained 8.7%, overstating it about sixfold.

E. Plausibility Versus Faithfulness

H1 is partly supported (Table V): plausibility and faithfulness were never strongly correlated ($|\rho| \leq 0.54$), but $|\rho| < 0.3$ held in only 12 of 24 pairs of method and condition and the sign changed with what the model used. On the control model all six methods showed negative correlations; on atrophy models most were positive; on the shortcut model Grad-CAM and Grad-CAM++ were again negative. Fig. 1 summarises plausibility and correctness across conditions.

F. Shortcut Strength

At amplitude 0.3 the model preferred atrophy at every correlation (Table VI): moving the shortcut changed at most 0.4% of decisions, even at $p = 1.0$. Its dependence on the bump nevertheless grew with p, the logit shift rising from 0.03 to 0.10 to 0.25, and LRP, Integrated Gradients, Gradient × Input and Occlusion followed this small rise monotonically. Grad-CAM and Grad-CAM++ stayed at their location baseline until $p = 1.0$, so on these models their shortcut scores carried no information about reliance. At amplitude 1.0 the model switched to the shortcut: moving it reduced accuracy from 0.781 to 0.224 and changed 70% of decisions. Every method followed. Shortcut lift rose to 40.2 for Occlusion and 26.8 for LRP, atrophy lift fell to 0.87–1.00 and hippocampal enrichment to 0.31–0.91, below chance for every method. These maps look wrong by the plausibility criterion, yet they correctly describe a model that has stopped using the hippocampus; H2 is again not supported.

G. Randomisation Tests

Re-initialising only the output layer already decorrelated the maps of LRP, Integrated Gradients and Occlusion from the original, and after further stages they stayed at $\rho \leq 0.24$ (Table VII). Grad-CAM decorrelated from layer3 on, but 22–49% of its maps were constant once layer4 was randomised and are excluded from ρ. Grad-CAM++ failed: a random output layer left its map almost unchanged ($\rho = 0.95$), and it stayed at 0.59–0.71 through layer1. Gradient × Input decorrelated only

TABLE II
Classification over 5 folds (581 subjects) and shortcut reliance. Logit shift: mean change of the logit towards the class the bump indicates when the bump is moved.

Condition	Test AUC	Accuracy	Acc., shortcut moved	Logit shift
Resampling control	0.515 ± 0.036	0.484	–	–
A: atrophy only	0.992 ± 0.006	0.943	–	–
C: shortcut only	0.826 ± 0.070	0.747	0.253	6.46
B: atrophy + shortcut, $p = 0.9$	0.980 ± 0.025	0.924	0.921	0.10

TABLE III
Condition A, 581 subjects: mean [95% bootstrap CI]. Plausibility and correctness are relative to a random map; the last column is faithfulness on the control model, which learned nothing.

Method	Faithfulness	Plausibility	Correctness	Faithfulness (control)
Occlusion	0.859 [0.848, 0.871]	6.09 [5.80, 6.38]	3.17 [3.03, 3.31]	0.695
LRP	0.617 [0.595, 0.638]	2.14 [2.06, 2.23]	1.82 [1.77, 1.86]	0.465
Integrated Gradients	0.558 [0.533, 0.583]	1.40 [1.36, 1.43]	1.58 [1.55, 1.60]	0.408
Gradient × Input	0.546 [0.521, 0.571]	1.27 [1.23, 1.30]	1.57 [1.55, 1.60]	0.400
Grad-CAM	0.301 [0.276, 0.327]	1.14 [1.12, 1.16]	1.32 [1.26, 1.39]	0.053
Grad-CAM++	0.292 [0.265, 0.318]	1.14 [1.13, 1.15]	1.25 [1.20, 1.30]	0.072
Random map	0.013 [−0.013, 0.039]	1.00 [1.00, 1.00]	1.00 [1.00, 1.00]	−0.006

TABLE IV
Correctness against the shortcut location (AUPRC lift). A: model that never saw a shortcut (location baseline); B: model that barely uses it; C: model that relies on it. Last column: B's excess over the baseline as a share of C's.

Method	A (baseline)	B	C	B excess, % of C
Occlusion	0.93 [0.91, 0.94]	3.41 [3.03, 3.81]	29.62 [27.62, 31.75]	8.7%
LRP	0.87 [0.87, 0.88]	1.33 [1.27, 1.39]	17.62 [16.35, 18.99]	2.7%
Integrated Gradients	0.90 [0.90, 0.91]	1.07 [1.05, 1.08]	8.17 [7.53, 8.85]	2.3%
Gradient × Input	0.88 [0.87, 0.88]	1.01 [1.00, 1.03]	8.27 [7.63, 8.97]	1.8%
Grad-CAM	2.68 [2.48, 2.87]	2.53 [2.25, 2.84]	7.36 [6.58, 8.23]	−3.2%
Grad-CAM++	3.04 [2.83, 3.26]	2.67 [2.45, 2.90]	8.19 [7.36, 9.04]	−7.3%

TABLE V
Spearman ρ between per-volume plausibility and faithfulness (581 subjects per condition).

Method	Control	A	C	B ($p = 0.9$)
Occlusion	−0.49	+0.37	+0.18	+0.54
LRP	−0.36	+0.28	+0.25	+0.44
Integrated Gradients	−0.25	+0.16	−0.24	+0.15
Gradient × Input	−0.25	−0.02	+0.19	−0.12
Grad-CAM	−0.05	+0.32	−0.41	+0.33
Grad-CAM++	−0.39	+0.34	−0.50	+0.39

partly, to a plateau near 0.43, because multiplying by the input carries image structure into the map whatever the weights. Data randomisation agreed: under the control model, whose labels are unrelated to the images, maps of the same scans correlated with the originals at $\rho = 0.57$ for Grad-CAM++ and 0.44 for Gradient × Input, against at most 0.24 for the others. SSIM was high for Integrated Gradients, Gradient × Input and LRP despite low rank correlation because their maps follow the brain outline; as in [13], rank correlation is the primary measure. H4 is supported.

H. Occlusion Window

Occlusion's correctness against atrophy hardly depended on its window (3.09, 2.98, 3.50 at 8^3, 16^3 and 32^3; Table VIII), so its lead on that axis is not a product of coarse maps. Plausibility varied from 3.55 to 6.43 and peaked at the default window: the same model and scans give hippocampal concentrations differing by almost a factor of two depending on a setting of the method. No window favoured the shortcut's position in the atrophy model. On model B the shortcut lift was 3.21 at the default window but 1.92 and 1.90 at the others, so much of Occlusion's overstatement of B's shortcut dependence comes from the default window.

V. Discussion and Limitations

Faithfulness and correctness agreed: on atrophy the six methods ranked identically on both. Plausibility did not: it exaggerated differences between methods, was at

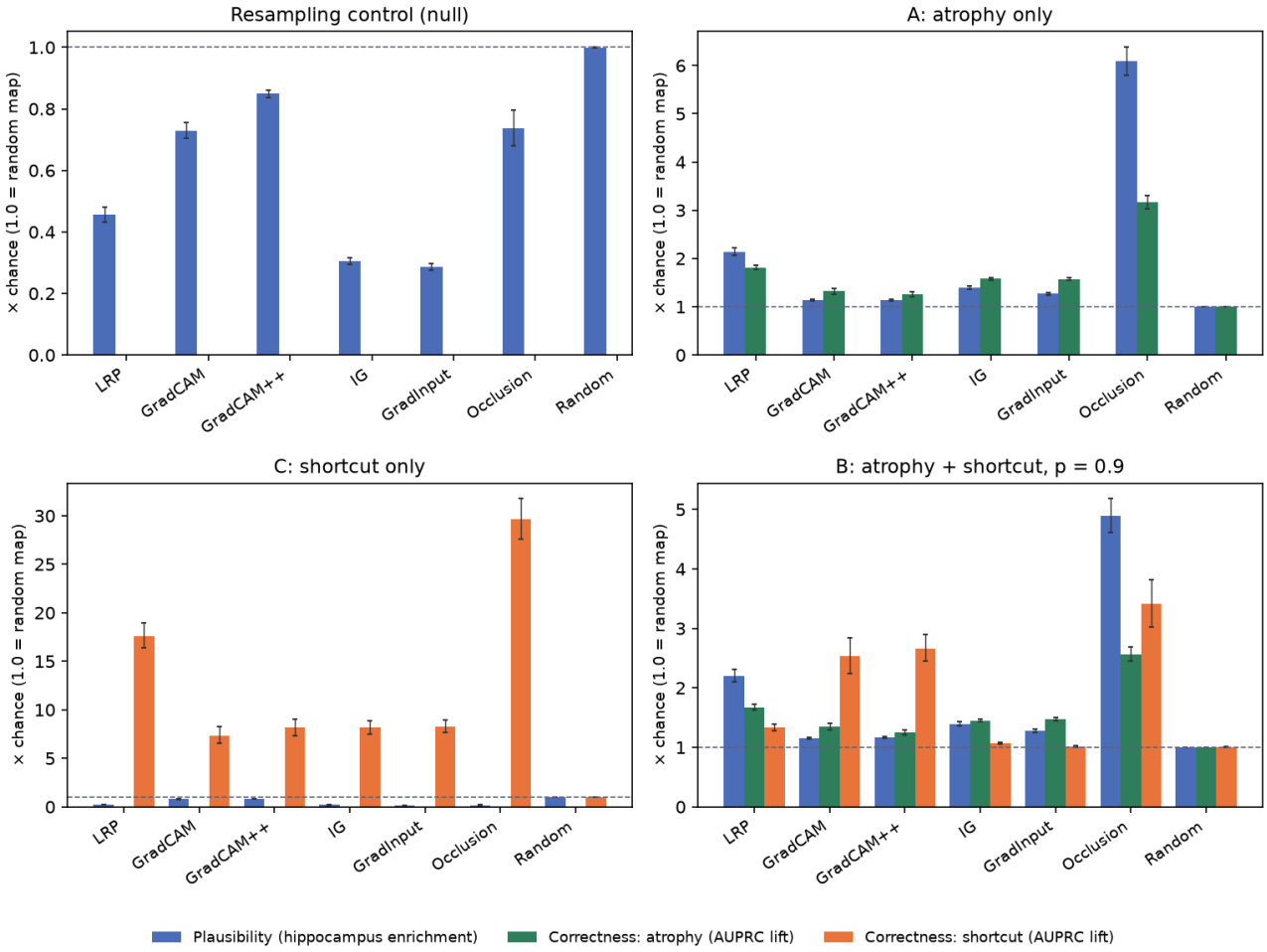


Fig. 1. Plausibility (hippocampus enrichment) and correctness (AUPRC lift against the atrophy and shortcut ground truth) per method and condition, relative to a random map (dashed line at 1.0). Error bars: 95% bootstrap CI over 581 subjects. Shortcut lift on model B should be read against the location baseline in Table IV.

most moderately correlated with faithfulness, and changed sign between conditions. Faithfulness was necessary but not sufficient, being almost as high on a model that had learned nothing. Correctness against ground truth was the only axis that separated these cases, and it too required a control: Grad-CAM and Grad-CAM++ favour a particular brain position strongly enough to appear to detect a shortcut placed there. Benchmarks that plant a signal at a fixed location should therefore report a no-signal baseline at that location. LRP, once its conservation error was corrected, was the best gradient-based method on atrophy correctness and, with Integrated Gradients and Gradient × Input, attributed relevance to the shortcut in proportion to the model's actual dependence on it across the sweep, and every method followed a model that switched to the shortcut. The randomisation tests separate the methods further: Grad-CAM++ maps barely depend on the model and Gradient × Input maps only partly. Plausibility is also sensitive to the method's own settings: Occlusion's window changed its hippocampal concentration almost twofold while its correctness stayed the same.

Each fold was trained once with one seed, so the intervals reflect variation between subjects, not training runs. The sweep produced models that barely use the shortcut or switch to it, but none that uses both signals in comparable measure. The randomisation tests and the window ablation used 100 subjects (20 per fold), and ablations of the IG baseline, the LRP residual split and atrophy severity remain to be run. The location control was added after a pilot run and is exploratory. Simulated atrophy is not Alzheimer's disease: it reproduces hippocampal volume loss but not cortical thinning, ventricular change or disease heterogeneity, so our claims concern explanation methods, not diagnosis. Future work covers intermediate shortcut amplitudes, the remaining ablations and a qualitative check on real AD scans [17].

TABLE VI

Shortcut sweep, 581 subjects per set (amplitude 0.3 unless stated): accuracy with the shortcut in place and moved, logit shift towards the bump's class, and shortcut correctness (AUPRC lift) per method. The first row is the location baseline of the atrophy-only model. Occ.: Occlusion; IG: Integrated Gradients; G×I: Gradient × Input; GC: Grad-CAM.

Condition B	Accuracy	Logit shift	Occ.	LRP	IG	G×I	GC	GC++
Location baseline	–	–	0.93	0.87	0.90	0.88	2.68	3.04
$p = 0.7$	$0.916 \rightarrow 0.916$	0.03	1.21	0.99	0.96	0.91	2.69	3.06
$p = 0.9$	$0.924 \rightarrow 0.921$	0.10	3.41	1.33	1.07	1.01	2.53	2.67
$p = 1.0$	$0.900 \rightarrow 0.898$	0.25	11.82	2.46	1.34	1.26	3.90	5.26
$p = 0.9$, amplitude 1.0	$0.781 \rightarrow 0.224$	8.23	40.22	26.81	13.28	13.25	6.07	7.12

TABLE VII

Randomisation tests on the atrophy models, 100 subjects (20 per fold). Cascading: Spearman ρ of the absolute map with the original after re-initialising the network from the output layer (fc) down to the named stage. Data: similarity to the map under the control model, whose labels are unrelated to the images. Pass: near zero.

Method	fc	layer4	layer3	layer2	layer1	stem	Data ρ	Data SSIM
Occlusion	−0.27	0.18	0.12	0.14	0.13	0.14	0.24	0.10
LRP	−0.25	0.15	0.10	0.11	0.10	0.10	0.14	0.71
Integrated Gradients	−0.11	0.24	0.15	0.16	0.16	0.15	0.17	0.89
Gradient × Input	0.17	0.47	0.43	0.43	0.42	0.43	0.44	0.89
Grad-CAM	0.21	0.37	0.00	−0.01	−0.05	−0.04	0.10	0.05
Grad-CAM++	0.95	0.67	0.59	0.70	0.71	0.29	0.57	0.50

TABLE VIII

Occlusion window (stride half the window), 100 subjects. Plaus.: hippocampus enrichment; Faith.: insertion − deletion; Atr.: atrophy lift; Loc.: lift at the shortcut location in the atrophy model; SC (B): shortcut lift on model B.

Window	Plaus.	Faith.	Atr.	Loc.	SC (B)
8^3	4.15	0.86	3.09	0.90	1.92
16^3 (default)	6.43	0.87	2.98	0.91	3.21
32^3	3.55	0.69	3.50	0.99	1.90

VI. Conclusion

A ground-truth benchmark makes the faithfulness and correctness of 3D MRI explanations measurable rather than assumed. Faithfulness and correctness agreed on how to rank six methods, plausibility did not, and faithfulness was high even for a model that had learned nothing. A no-signal control showed that Grad-CAM's apparent detection of a shortcut came from where the shortcut was placed, every method followed a model that switched to a stronger shortcut, and randomisation tests showed that Grad-CAM++ maps barely depend on the model. Plausible-looking maps are not evidence of correct ones.

Acknowledgment

This study uses only the publicly available, anonymised IXI dataset (https://brain-development.org), released under CC BY-SA 3.0; no new data were collected. The code and all results are available at https://github.com/UdayBhardwaj/alz.

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